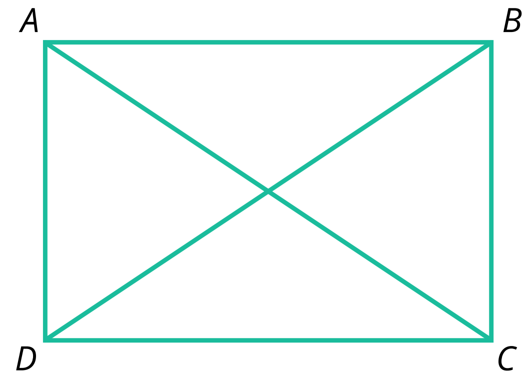
Properties of rectangles, rhombuses, [redacted] squares

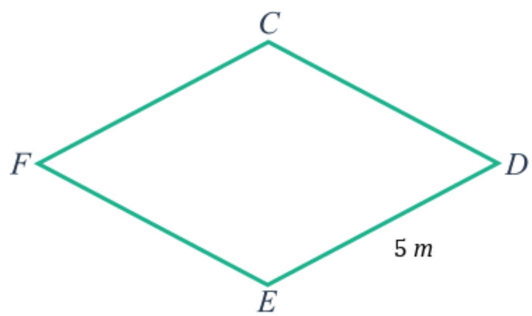
1 Consider the given rectangle $ABCD$.
Suppose that $\overline{AC} = 16 \text{ m}$. Find:

a $\overline{BD}$ **b** $m\angle BCD$

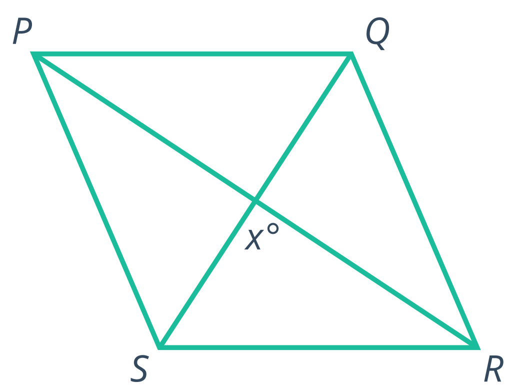


2 Consider the given rhombus $CDEF$.
Find:

a $\overline{CD}$ **b** $\overline{EF}$ **c** $\overline{FC}$



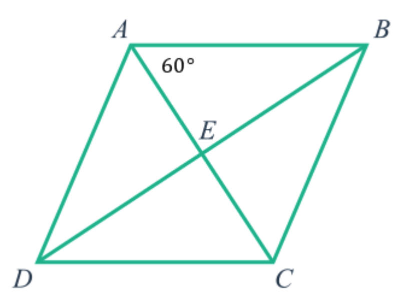
3 Consider the given rhombus $PQRS$.
Find the value of x.



4 Consider the given rhombus $ABCD$.
Find:

a $m\angle BEC$ **b** $m\angle BCE$

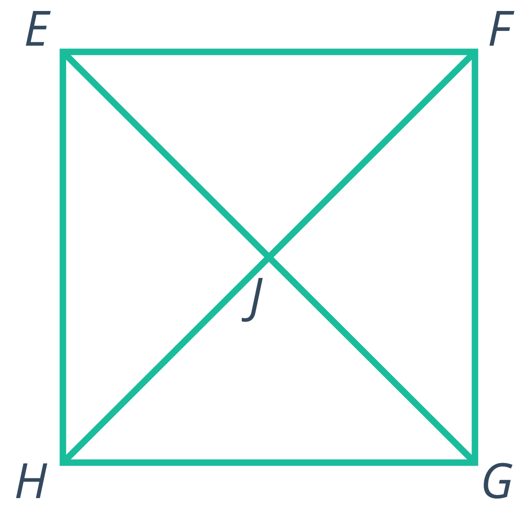
c $m\angle ECD$ **d** $m\angle BCD$



5 Consider the given square $EFGH$.
Suppose that $JG = 19 \text{ cm}$. Find:

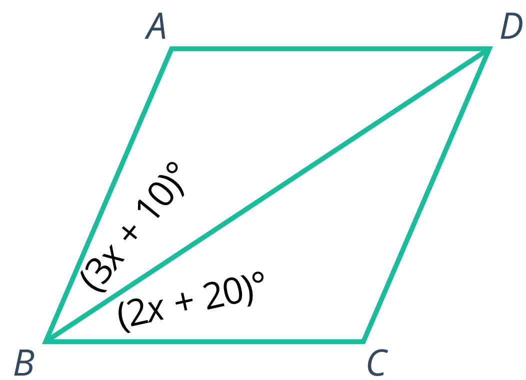
a $\overline{EJ}$ b $\overline{FH}$

c $m\angle EJF$ d $m\angle JFG$



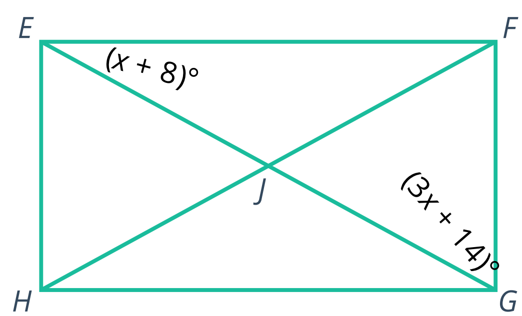
6 Consider the given rhombus $ABCD$.

a Solve for x. b Find $m\angle ABD$.



7 Consider the given rectangle $EFGH$.

a Solve for x. b Find$m\angle HJG$.



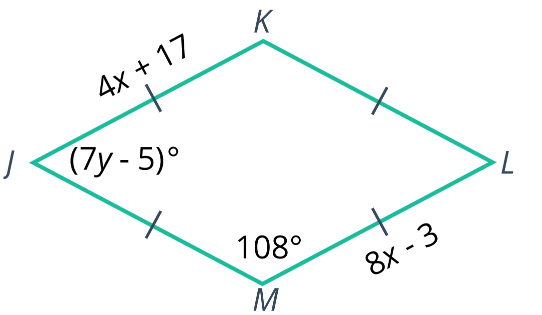
8 For each of the following parallelograms:

i Classify the parallelogram as rectangle, rhombus, or square.

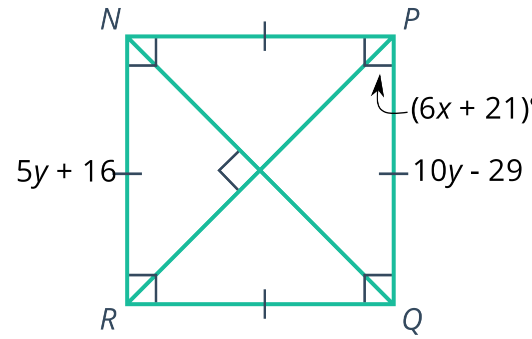
ii Solve for x

iii Solve for y

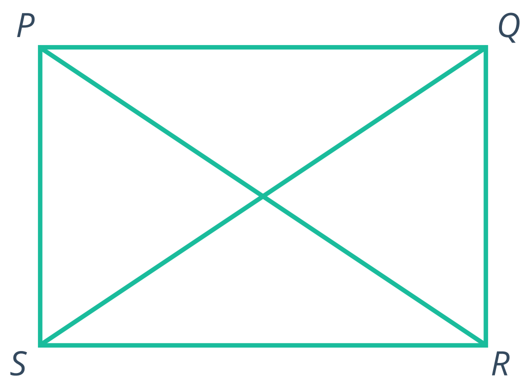
a



b



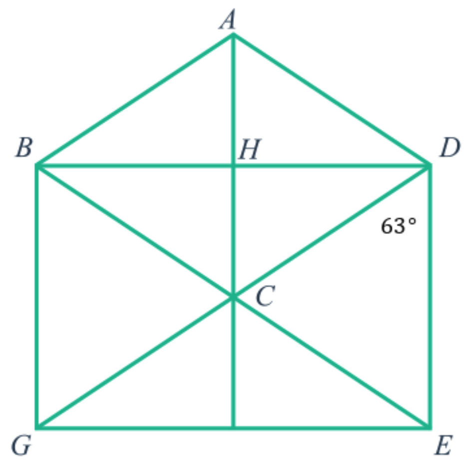
9 In the given rectangle $PQRS$:
$\overline{PQ} = x + 50;$
$\overline{PR} = 2x + 10;$ and
$\overline{SQ} = 4x - 130$



a Solve for x.

b Find the length of $\overline{PQ}$.

10 In the given diagram, $BDEG$ is a rectangle and $ABCD$ is a rhombus.
Find:

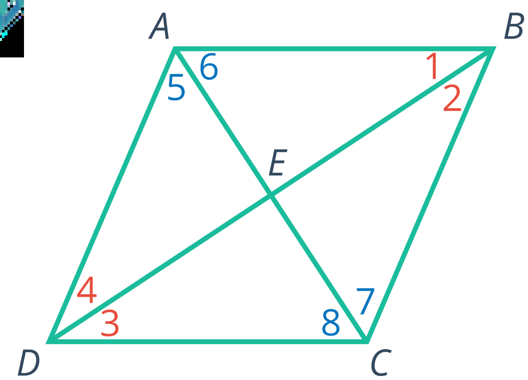


a $m\angle BDE$ b $m\angle BDC$

c $m\angle ADC$ d $m\angle BAD$

e $m\angle CAD$ f $m\angle ACD$

11 Consider the following proof based on the given rhombus, $ABCD$:



To prove: $\angle 1 \cong \angle 2 \cong \angle 3 \cong \angle 4 \cong \angle 5 \cong \angle 6 \cong \angle 7 \cong \angle 8$

	Statements	Reasons
1.	$ABCD$ is a rhombus	Given
2.	$ABCD$ is a parallelogram $\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$	Definition of a rhombus
3.	$\overline{AE} \cong \overline{EC}$ and $\overline{DE} \cong \overline{EB}$	If a quadrilateral is a parallelogram, then its diagonals bisect each other.
4.	$\overline{AE} \cong \overline{AE}, \overline{BE} \cong \overline{BE}$, $\overline{CE} \cong \overline{CE}$, and $\overline{DE} \cong \overline{DE}$	Reflexive property of congruence
5.	$\triangle AEB \cong \triangle CEB \cong \triangle CED \cong \triangle AED$	Side-side-side congruence theorem
6.	$\angle 1 \cong \angle 2 \cong \angle 3 \cong \angle 4 \cong \angle 5 \cong \angle 6 \cong$ $\angle 7 \cong \angle 8$	Corresponding parts of congruent triangles are congruent (CPCTC).

Determine whether each statement is true or false.

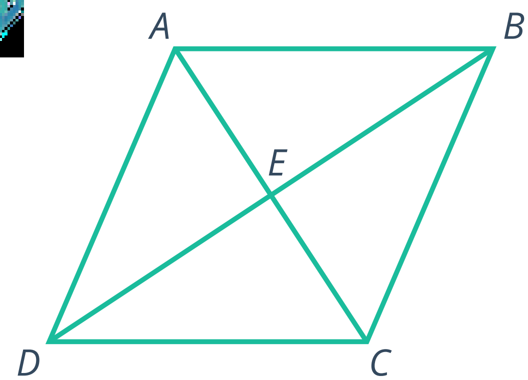
a If a quadrilateral is a rhombus, then its diagonals bisect the angles at the vertices.

b If a quadrilateral is a rhombus, then its diagonals are congruent.

c If a quadrilateral has diagonals which bisect the angles at the vertices, then it is a rhombus.

d If a quadrilateral is a parallelogram, then its diagonals bisect the angles at the vertices.

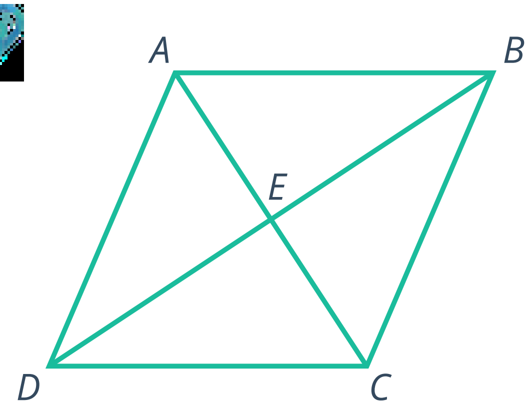
12 Complete the following proof to show that the given rhombus $ABCD$ is composed of four congruent triangles:



To prove: $ABCD$ is composed of four congruent triangles

	Statements	Reasons
1.	$ABCD$ is a rhombus	Given
2.	$ABCD$ is a parallelogram $\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$	Definition of a rhombus
3.	▯	▯
4.	$\overline{AE} \cong \overline{AE}, \overline{BE} \cong \overline{BE}$, $\overline{CE} \cong \overline{CE}$, and $\overline{DE} \cong \overline{DE}$	Reflexive property of congruence
5.	$\triangle AEB \cong \triangle CEB \cong \triangle CED \cong \triangle AED$	Side-side-side congruence theorem

13 Consider the following proof based on the given rhombus, $ABCD$:



To prove: $\overline{AC} \perp \overline{BD}$

	Statements	Reasons
1.	$ABCD$ is a rhombus	Given
2.	$\triangle AEB \cong \triangle CEB \cong \triangle CED \cong \triangle AED$	If a quadrilateral is a rhombus, then it is composed of four congruent triangles.
3.	$\angle AEB \cong \angle AED$ (and two other angles in the center)	Corresponding parts of congruent triangles are congruent (CPCTC).
4.	$\angle AEB$ and $\angle AED$ form a linear pair (and two other angles in the center)	Given
5.	$\angle AEB$ and $\angle AED$ are right angles (and two other angles in the center)	Two congruent angles that form a linear pair are right angles.
6.	$\overline{AC} \perp \overline{BD}$	Definition of perpendicular

Determine whether each statement is true or false.

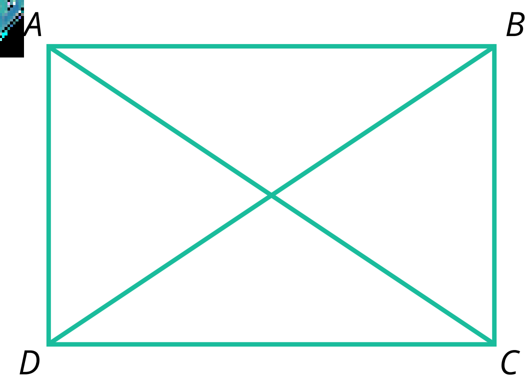
a If a quadrilateral is a rhombus, then its diagonals are perpendicular.

b If a quadrilateral is a parallelogram, then its diagonals bisect the angles at the vertices.

c If a quadrilateral is a rhombus, then its diagonals are congruent.

d If a quadrilateral has diagonals that are perpendicular, then it is a rhombus.

14 Complete the following proof to show that in the given rectangle $ABCD$, $\overline{AC} \cong \overline{BD}$:



To prove: $\overline{AC} \cong \overline{BD}$

	Statements	Reasons
1.	$ABCD$ is a rectangle	Given
2.	$ABCD$ is a parallelogram $\angle ADC$ and $\angle BCD$ are right angles	Definition of a rectangle
3.	$\angle ADC \cong \angle BCD$	All right angles are congruent.
4.	$\overline{AD} \cong \overline{BC}$	If a quadrilateral is a parallelogram, then its opposite sides are congruent.
5.		
6.	$\triangle ACD \cong \triangle BDC$	Side-angle-side congruence theorem
7.	$\overline{AC} \cong \overline{BD}$	Corresponding parts of congruent triangles are congruent (CPCTC).

Additional questions

15 Determine whether each statement is true or false:

a All quadrilaterals are parallelograms

b All rectangles are parallelograms

c All parallelograms are rhombi

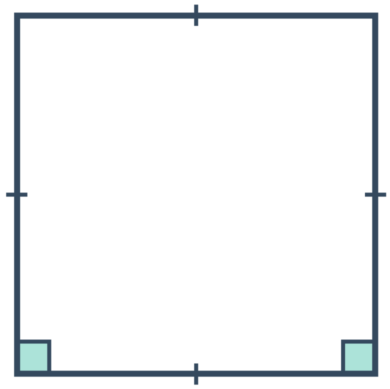
d A parallelogram with four congruent sides is a square

e A rectangle with consecutive sides congruent is a square

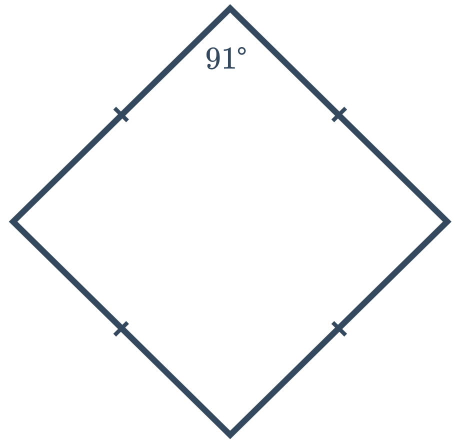
f A rhombus is a parallelogram

16 Classify each quadrilateral. Explain your rea[redacted]g.

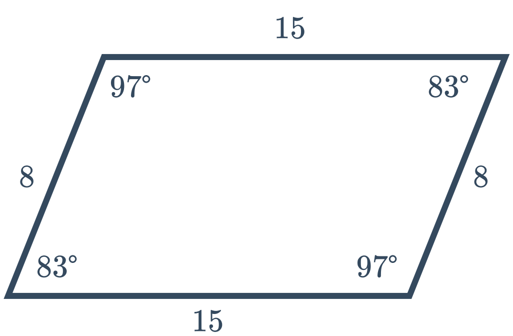
a



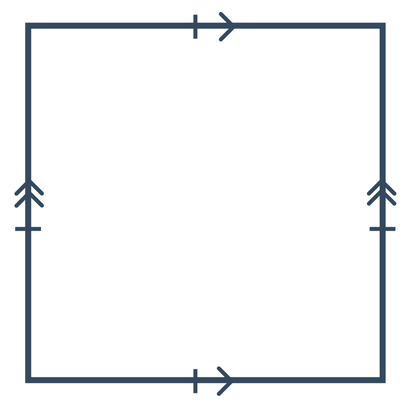
b



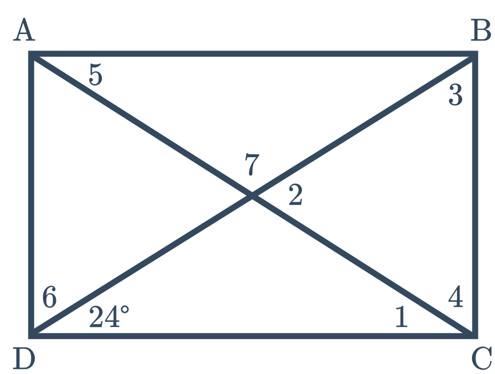
c



d

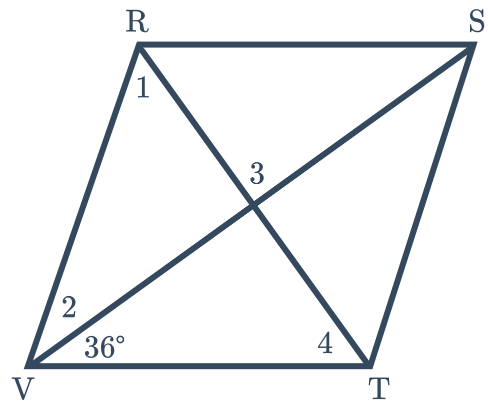


17 Given: $ABCD$ is a rectangle
Solve for each missing angle in the diagram.



18 Given: $RSTV$ is a rhombus

Solve for each missing angle in the diagram.

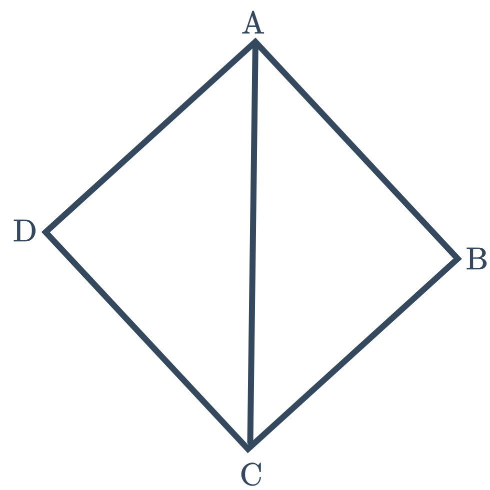


19 Given: ABCD is a rhombus

$m\angle DAC = 4x + 9$

$m\angle DAB = 11x - 3$

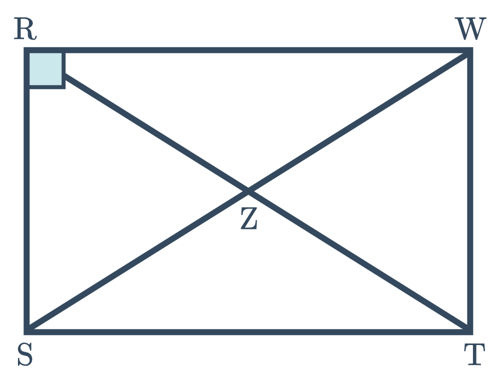
- **a** Solve for x
- **b** Solve for $m\angle BAC$
- **c** Solve for $m\angle ABC$



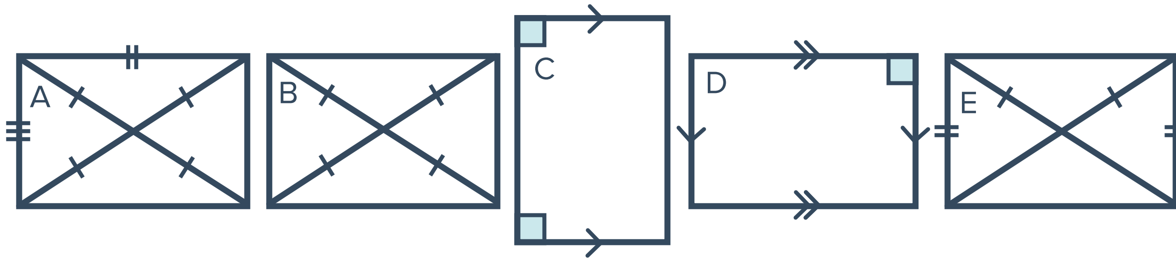
20 $RZ = 4x + 10$

$SW = 5x - 20$

- **a** Classify the quadrilateral.
- **b** Solve for x.



21



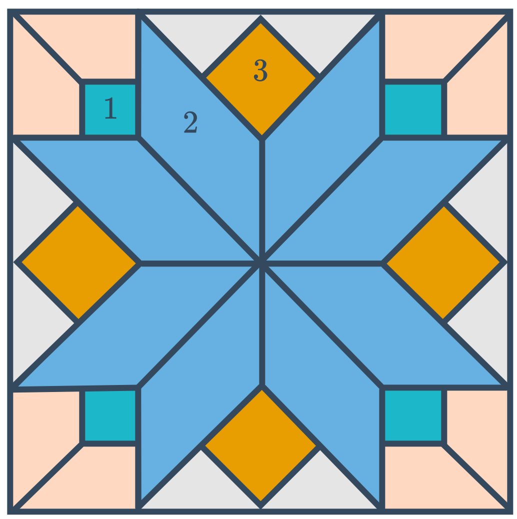
a Based on the given markings, which of the following quadrilaterals is not a rectangle? Explain how you know.

b What additional information could make the quadrilateral you found in part (b) a rectangle?

22 Stained glass windows are made by putting together various shapes, like those in the design below:
For quadrilaterals 1-3 in the design:

a Make your best guess as to the classification of the quadrilateral.

b Determine whether you could correctly classify the quadrilaterals using only a ruler.



23 Sketch each of the following, labeling all sides and angles accordingly:

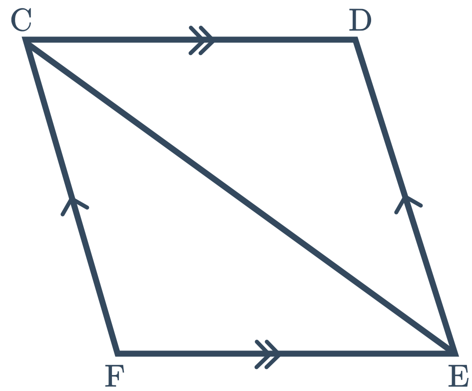
a A parallelogram that is neither a rectangle or a rhombus.

b A parallelogram with a pair of consecutive congruent sides and a pair of acute angles.

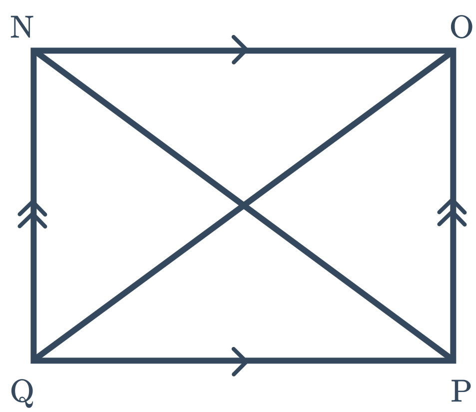
c A parallelogram with a right angle.

24 Determine whether the given information is enough to prove the given statement. Justify your answer.

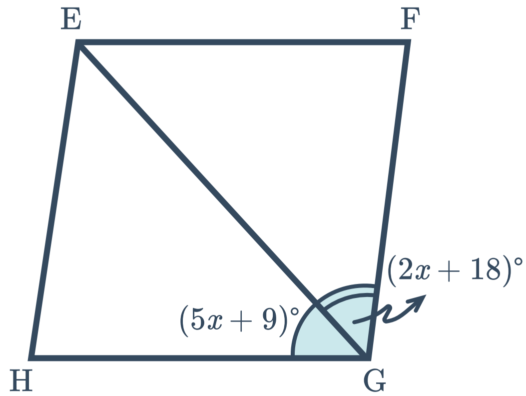
a Given: Parallelogram $CDEF$
Conclusion: $CDEF$ is a rhombus.



b Given: Parallelogram $NOPQ$
$\overline{NO} \cong \overline{PQ}$
Conclusion: $NOPQ$ is a rectangle.



c Given: Parallelogram $EFGH$
$x = 3$
Conclusion: $EFGH$ is a rhombus.



25 If $\overline{AB} \cong \overline{CD}$, show that $ABCD$ is not a rhombus.